The International Journal of Management Education 22 (2024) 100927

Contents lists available at ScienceDirect

The International Journal of Management Education

journal homepage: www.elsevier.com/locate/ijme

Examining different learning modes: A longitudinal study of business administration students' performance

Islam Elbayoumi Salem [a,b,*], Alamir AL-Alawi [c], Sanyo Moosa [a], Lujain El-Maghraby [d], Nasser Alhamr Alkathiri [a], Ahmed Mohamed Elbaz [a,e]

[a] University of Technology and Applied Sciences-Salalah, College of Economics and Business Administration, Department of Economics and Business Administration, Sultanate of Oman
[b] Alexandria University, Faculty of Tourism and Hotels, Department of Hotel Management, Alexandria, Egypt
[c] University of Technology and Applied Sciences, Ibri, Oman
[d] Department of Management, College of Commerce and Business Administration, Dhofar University, PO Box: Salalah 2509, 211, Salalah, Oman
[e] University of Sadat City, Faculty of Tourism and Hotels, Tourism Studies Department, Egypt

ARTICLE INFO

Keywords:
Learning satisfaction
Motivation
Perceived ease of use
Perceived usefulness
Student performance
Blended learning
Hybrid learning
Online education and face-to-face education

ABSTRACT

This study examines how different learning modes (blended learning, online learning, and face-to-face) affect business administration student performance. Four dimensions are examined: perceived usefulness, perceived ease of use, learning motivation, and students' satisfaction. The current study used longitudinal data from three survey copies that were distributed to the same student to compare face-to-face, online, and blended modes over three different time periods (2019, 2020, and 2021). Participants included 160 business students at local universities. The results indicated that students' motivation to use face-to-face learning had a great impact on their satisfaction, but this had only a marginal effect on their academic performance. On the other hand, students' motivation to use online learning significantly influences their learning satisfaction and, in turn, has a strong impact on their academic performance. As for the blended learning model, students' motivation to use this mode of learning positively impacted their learning satisfaction. However, students of this mode had relatively lower academic performance compared to other modes of learning.

1. Introduction

Over the past few years, higher education has been rapidly influenced by the ever-changing developments in information and communications technology (ICT) (Truss & Anderson, 2023). This has redefined learning and teaching processes, in particular online education. The rise of online education has significantly altered the contours of teaching and learning as a result of this technological boom. The onset of the COVID-19 pandemic has further accelerated the adoption of e-learning, sparking a profound shift in educational paradigms (Fang, Pechenkina, & Rayner, 2023; Mali & Lim, 2021). As a response to the unprecedented challenges posed by the pandemic, traditional face-to-face instruction gave way to an overwhelming surge in e-learning across higher education institutions

* Corresponding author. College of Economics and Business Administration, University of Technology and Applied Sciences, Salalah, Oman.
E-mail addresses: islam@utas.edu.om, islam.salem@alexu.edu.eg (I.E. Salem), alameer.alalawi@utas.edu.om (A. AL-Alawi), sanyo@utas.edu.om (S. Moosa), Lelmaghraby@du.edu.om (L. El-Maghraby), nasser.m.alkatheri@utas.edu.om (N.A. Alkathiri), ahmed.elbaz@utas.edu.om, ahmed.elbaz@fth.usc.edu.eg (A.M. Elbaz).

https://doi.org/10.1016/j.ijme.2023.100927
Received 28 March 2023; Received in revised form 29 November 2023; Accepted 19 December 2023
Available online 29 December 2023
1472-8117/© 2023 Elsevier Ltd. All rights reserved.

worldwide. This transition triggered fervent discussions among scholars, who fervently debated the merits and demerits of this emergent educational paradigm (Singh, Steele, & Singh, 2021; Kumar, 2021). Blended learning, which combines online with face-to-face elements to complement the learning process on campus, has been utilised as another mode (Dhawan, 2020). Prior work mentions that this mode presents an interactive learning experience with flexibility in using learning materials and increases discussion between students and faculty members either on a one-to-one basis or in small teams (Prohorets & Plekhanova, 2015). In addition, blended learning has the potential to change students' experiences and outcomes through learning (Peimani & Kamalipour, 2022). KK & Maskari (2019), in their study, examined the students' perceptions of blended learning instructional design in a higher education institution. They concluded that there is a positive perceived attitude among students towards blended learning courses, in particular with respect to learner autonomy, student engagement, practical learning that lessens the gap between academia and industry and learning flexibility. Another advantage of blended learning environments is their potential to offer many sources for learners (Peimani & Kamalipour, 2022). Bondarenko, Mantulenko, & Pikilnyak (2019) concluded that blended learning provides extra resources for learners, which in turn may improve learners' confidence and proficiency as well as student performance. Maulida, Rahman, Handrianto, and Rasool (2022) outlined other advantages of blended learning, such as a deep understanding of topics by using web-based resources as well as the active participation of students in class. In addition, online learning engagement may enable team and cooperative activities even beyond the classroom experience itself (Czaplinski & Fielding, 2020), as well as their critical thinking abilities (Yean, 2019).

However, within this evolving educational landscape, there exists a critical research gap pertaining to learner satisfaction across three distinct learning modes: online, hybrid, and on-campus education. Furthermore, the factors directly or indirectly influencing students' learning satisfaction within these varied learning environments remain under-explored. In this regard, past researchers of learning have called for studying the relationship between perceived ease of use, perceived usefulness, learning motivation, learning satisfaction, and students' performance (Huenag, 2021; Chen, Heng, Chye, Seng, & Yuan, 2022; Daud, Liting, Nasir, Hadini, & Bahri, 2022). Accordingly, this study aims to fill this research void by developing and testing a conceptual model to compare the factors that determine student performance in the higher education context through these three educational delivery techniques. This was based on the theoretical lens of the Technology Acceptance Model (TAM). The study investigated these types from the perspectives of perceived ease of use, perceived usefulness, learning motivation, and student satisfaction. By doing so, this study contributes to the theory of student performance by establishing a conceptual model of the factors that determine student performance in a higher education context from the perspectives of perceived ease of use, perceived usefulness, learning motivation, and student satisfaction. To describe the structure model and investigate the interrelationship among determining factors, the partial least squares method (PLS-SEM) has been used. This study shall provide empirical evidence for policymakers, practitioners, and academicians about important factors that can better improve learning satisfaction and, hence, students' performance by addressing the following questions:

1. How does perceived ease of use impact perceived usefulness in the context of face-to-face, online, and blended learning modes?
2. What is the influence of perceived ease of use on students' motivation within the realms of face-to-face, online, and blended learning environments?
3. How does perceived usefulness affect students' motivation across face-to-face, online, and blended learning modes?
4. To what extent does learning satisfaction contribute to student performance in the contexts of face-to-face, online, and blended learning modalities?

2. Theoretical background

2.1. Technology accepted model (TAM)

The current study investigated the impact of perceived ease of use and perceived usefulness on student motivation, satisfaction, and performance in three primary learning modalities: face-to-face, online, and blended learning. We are also examining this relationship using the theoretical framework of the Technology Acceptance Model (TAM). Fishbein and Ajzen (1975) proposed the theory of reasoned action, which established the fundamental principles for building the technology acceptance model (TAM). TAM aims to elucidate the factors influencing consumers' inclination to embrace novel technologies. The main objective of the Technological Acceptance Model (TAM) was to clarify the fundamental mechanisms that influence the adoption of technology in a higher education setting. The purpose of this was to predict the actions of people and establish a theoretical structure to explain the successful incorporation of technology. The primary factors in this model are the perceived usefulness (PU) and the perceived ease of use (PE), which have an impact on user intentions to adopt technology (Hanafizadeh & Shaikh, 2021; Karjaluoto, Shaikh, Saarijärvi, & Saraniemi, 2019; Shaikh, Glavee-Geo, Karjaluoto, & Hinson, 2023). Perceived ease of use is the level of awareness a person has regarding the ease of using a specific system, as defined by Huang (2021). On the other hand, perceived usefulness relates to the level of trust a user has in new technology that can improve their work performance, as described by Fuady, Sutarjo, and Ernawati (2021). In this vein, the primary aim of our research is to examine the impact of several learning modes (blended learning, online learning, and face-to-face) on the performance of business administration students in terms of perceived utility, perceived ease of use, learning motivation, and student satisfaction. We chose TAM specifically because it aligns with the dimensions, we are investigating in our research topic. Another reason is that the Theory of Planned Behaviour (TAM) is widely acknowledged as the predominant theory when it comes to the four dimensions it examines, which are of utmost significance to our study.

2.2. Higher education sector in Oman

Oman is an integral part of the world's economy after signing the General Agreement on Trade in Services in 2002. Therefore, the country has adopted serious measures to renovate the higher education sector in terms of quantity and quality. Higher education is perceived in the country as the major pathway to developing a world-class human resources capacity for the workforce. Internationalisation has been a motivation for Oman to focus on higher education to best address financial developments in the international market (Donn & Issan, 2007). Amid pervasive regional political flux and reducing oil prices, Omani officials have initiated reforms and strategic initiatives in higher education (Al-Abri, 2015). Hence, the Omani higher education system accommodates local, regional, and international standards and demands for education. Yet, many studies blame higher education in Oman for not being able to satisfy the needs of the public and private sectors in the labour market. IT skills and critical thinking skills were among the most in short supply (Baporikar & Shah, 2012). Al-Harthi (2002) complained that the education system can produce a power-distance relationship inside classrooms, whereby educators have solitary authority. The scholar blamed this system of rigid hierarchy in classrooms for discouraging the motivation of students to exercise critical thinking in their academic studies. Under these conditions, it is improbable to come up with the right academic environment that fosters motivation and student performance. A more engaging classroom context is necessary to enhance student performance.

2.3. Blended learning, online learning and traditional learning

Education can occur in multiple modes, often differentiating between on-campus classroom learning and online learning, as well as synchronous and asynchronous learning (Shahabadi & Uplane, 2015). Nowadays, distance education is fundamentally used for asynchronous learning, whereas combination with on-campus instruction is called hybrid, blended, or flexible learning (Müller & Mildenberger, 2021). Many scholars have examined blended learning from different perspectives, like the educational perspective (Rodríguez & Rojo, 2020), the holistic perspective (Gecer, & Dag, 2015), the pragmatic perspective (Bouguebs, 2019), the chief learning officer (CLO) perspective (Chen, 2022), and the corporate training perspective (Mahande, Darmawan, & Malago, 2021). Dimensions constituting hybrid learning are combining offline and online learning; collaborative learning; combining self-paced and live blending; and blending structured and unstructured learning (Kaur, 2013).

One major challenge is that blended learning is defined in different ways when compared to e-learning and face-to-face learning (Mueller & Wulf, 2022; Smith & Hill, 2019). The most cited Graham, Allen, and Ure (2005) stated three orientations: (a) blended learning as a blend of various learning or delivery media; (b) blended learning as multiple teaching methods; or (c) blended learning as a blend of face-to-face education with computer-facilitated instruction. Driscoll (2002) argues that four kinds of combinations can be depicted: (a) blending modes of computer-based technology to achieve an educational objective; (b) blending educational approaches to create an optimal learning outcome with or without pedagogical technology; (c) blending any type of pedagogical technology with on-campus training; and (d) blending pedagogical technology with practical job tasks (working and learning). Therefore, one can infer that hybrid learning constitutes all ICT-facilitated learning except for 100% virtual learning environments and 100% face-to-face instruction (Müller & Mildenberger, 2021).

To ensure the "right combination" between on-campus teaching and online education, So and Bonk (2010) argue that the following questions need to be addressed: Firstly, which component of the interaction should occur on campus or in virtual settings? Secondly, what are the appropriate circumstances for utilizing online or classroom-based face-to-face learning, and thirdly, how can these two modes be combined to optimize the effectiveness of the learning process? Allen, Seaman, and Garrett (2007) differentiate between the following learning modes based on the "online component proportion": face-to-face, ICT-supported, flexible/hybrid, and e-learning. According to Müller and Mildenberger (2021), hybrid learning entails an online proportion between 30 and 79 percent of curriculum delivery. On the contrary, other scholars suggest different proportions of online learning, for instance: a 1:1 ratio for classroom and online teaching (Bernard, Borokhovski, Schmid, Tamim, & Abrami, 2014), the "course and a half" syndrome (Garrison & Vaughan, 2008, p. 202), or the lower end of 30 percent, which emphasises that a significant portion of the teaching occurs virtually online (Allen et al., 2007). Anthony, Kamaludin, and Romli (2022) argue that academics utilize blended learning, which consists of online and face-to-face campus activities, resources, information, feedback, and assessment, to enhance learning satisfaction and performance. They concluded that effective blended learning is shaped by ease of use, which includes appropriate computer software and hardware resources, educational support, financial assistance, and faculty promotion aspects.

3. Hypotheses development

3.1. Perceived usefulness and perceived ease of use

Previous studies have conducted empirical investigations on the perception of ease of use and usefulness in the same research setting. In their study, Fuady et al. (2021) sought to elucidate the perceptions of students regarding the ease and usefulness of different learning media options, specifically Zoom, Google Meet, Google Classroom, and LMS. Participants reported Zoom as the easiest media learning platform, while considering LMS more complicated than the other three media. Interestingly, the same media platforms were reported again and in the same order when examining perceived usefulness. For example, Zoom was considered the most useful learning media platform in comparison to the other three methods, while LMS was considered less useful than the other three media. This suggests that perceived ease of use and perceived usefulness are highly interrelated. Bachtiar, Rachmadi, and Pradana (2014) tested the TAM model to examine the adoption of online learning as a complementary tool in a traditional learning context. System

Interactivity (SI), System Functionality (SF), Usability (U), Internet and Computer Experience (ICE), Self-Efficacy (SE), Perceived Ease of Use (PEU), Socio-Environmental Factor (SEF), Perceived Usefulness (PU), Use for Distance Learning (UDL), and Use of Supplementary Learning (USL) are the main factors that affect how the system is used. Davis (1989) posits that consumers are more likely to perceive a system or technology as useful when it is viewed as easy to use. Empirical evidence supports the use of TAM to test the acceptance of blended learning. They also showed that perceived usefulness (PU) is the key determinant of the construct, followed by the variables "perceived ease of use" and "use of supplementary learning" to depict the causal relationship in the TAM model. Hence, we hypothesize:

H1a. Perceived ease of use positively affects the perceived usefulness of face-to-face learning mode.

H1b. Perceived ease of use positively affects the perceived usefulness of online learning mode.

H1c. Perceived ease of use positively affects the perceived usefulness of the blended learning mode.

3.2. Student's motivation

Motivation to learn is an intrinsic and extrinsic inspiration to students who are learning to achieve changes in behaviour, typically pooling multiple indicators or supporting elements (Saiful et al., 2018). Motivation influences learning effectiveness significantly (Riswanto & Aryani, 2017). Learning motivation dimensions can be categorised as 1) drive to succeed; 2) support and need in education; 3) future prospects and ambitions; 4) obligation towards learning; 5) activities in learning stimulating interest; and 6) a supportive pedagogical context. Dwijuliani (2021). Educators opt for different mechanisms to increase students' motivation and their inner drive to learn (Kwon & Choi, 2022). Dwijuliani (2021) stated three strategies to ignite learners' motivation that academicians may implement. These include 1) introducing learning variations such that learners can stay focused and concentrated in the education process; 2) engaging students through active learning and problem-solving exercises; and 3) creating a conducive learning classroom environment.

Past studies have concluded that there are multiple elements that motivate students to learn courses online during the COVID-19 period. These include a sense of ownership (Oh, Chan, & Kim, 2020; Oe, Takemoto, & Ridwan, 2020). Asynchronous methods like e-portfolios and gamification-based modules can boost motivation levels among learners as they can exercise a sense of ownership towards their assignments. That is, students are able to use and take control of the whole learning system. They also find ease in doing research by themselves about the courses that they are learning (Oe et al., 2020). Moreover, the relevance of the curriculum materials used is also a factor in motivating the students, as it helps them understand the theme easily and fully appreciate the usefulness of the content being taught (Qunfei, Abdullah, & Mustapha, 2020; Famularsih, 2020). Students find it motivating when their teacher provides materials that are relevant to what they are learning. In addition, students also find it motivating when they can explore content that is of personal interest and relevance, as it makes it easier for them to comprehend the curriculum (Qunfei et al., 2020). Moreover, e-learning can boost active participation among students, opening the door for intellectual exchanges and idea creativity, which in turn enhances perceived usefulness (Susilawati, 2020). All of this is in line with the TAM theory, which stipulates that perceived ease of use influences perceived usefulness and is a key factor in learning acceptance. Along the same lines, Lin et al. (2010) examined multiple factors that motivate students to constantly use e-learning and blended learning with multimedia content, support, or a multimedia e-learning system (MMLES). This research was conducted from the perspective of the Technology Acceptance Model (TAM), which emphasised the role of users' motivation in technology. They concluded that factors that influence students' motivation to use MMLES are perceived usefulness, system characteristics, perceived ease of use, perceived enjoyment, confidence in their ability to use MMLES, and features of the learning media. Hence, we hypothesize:

H2a. Perceived ease of use positively affects perceived motivation face-to-face learning mode.

H2b. Perceived ease of use positively affects perceived motivation for online learning mode.

H2c. Perceived ease of use positively affects perceived motivation for blended learning mode.

H3a. Perceived usefulness positively affects motivation for face-to-face learning mode.

H3b. Perceived usefulness positively affects motivation for online learning mode.

H3c. Perceived usefulness positively affects motivation for blended learning mode.

3.3. Student satisfaction

Student satisfaction is a well-established implication of user acceptance, user motivation, IT characteristics, and system characteristics and is often used to measure learners' satisfaction in pedagogical studies (Bray, Aoki, & Dlugosh, 2008). Past research revealed a number of inhibitors that impede learners' satisfaction, which include financial problems, technical issues, and a lack of appropriate ICT skills (Bullen & Janes, 2007). Nagy (2018) studied the responsible factors behind students' video usage and their learning satisfaction pertinent to the usage of supplementary applications of learning videos available on Moodle in a Business Mathematics subject. He based his conceptual model on extending the Technology Acceptance Model (TAM) with the following key constructs: perceived usefulness, perceived ease of use, internet self-efficacy, and motivation, as explanatory variables of video usage. The outcomes in question were learner-teacher interaction, learning performance, and learner-learner interaction. Data were gathered from

89 students using a questionnaire, adopting a partial least-squares structural equation modelling approach to test the hypothesis. The results confirmed that perceived usefulness, motivation, and internet self-efficacy had a direct effect on video usage. Perceived ease of use, learner-learner interaction, and learning performance directly impacted learning satisfaction. In addition, the findings showed that video usage had a considerable impact on learning performance and on learning satisfaction.

Howard and Maxwell (1980) contrasted the student motivation model with the grading leniency model. As per the student motivation model, satisfaction alone does not shape grades, but student motivation can act as an intermediary role in influencing both student satisfaction levels and grades. Their research revealed that satisfaction is associated more with student motivation and progress than with grades, signalling the fact that student satisfaction levels are not reduced by low grades. Wright (2017) examined students' perceptions of virtual lessons as contrasted with on-campus lessons. Questionnaires using quantitative (Likert scale questions) and qualitative (open-ended questions) methods revealed data for content analysis to identify common student perceptions, referring to motivation and interest. Wright (2017) also showed that more students' participation is correlated with on-campus lessons and with higher motivation and more attention because of better understanding, more enriching classroom interaction with the professor and colleagues, and efforts from the professor, which can be considered aspects of "perceived usefulness." Students preferring the virtual lesson mentioned that convenience, speed of study, and time-and-place flexibility are the reasons behind their choice, which can be considered as the perceived ease of use.

Zheng, Yu, and Wu (2022) pointed to the fact that contemporary media technology can support well-developed teacher-student rapport, which ignites students' motivation to learn and impacts their learning satisfaction. In this research, Chinese students befriended their instructors on social media platforms, and their blended learning (N = 455) based on this social media interaction was contrasted against face-to-face learning (N = 450). The findings indicate that affective learning and cognitive learning in blended learning ignited motivation and hence resulted in more student satisfaction compared to face-to-face learning. However, there were no significant differences in academic self-concept, social presence, or grade point average between traditional learning and blended learning. According to a multi-group structural equation modelling analysis, academic self-concept and motivation significantly influenced cognitive learning and, hence, satisfaction. Hence, we hypothesize:

H4a. Motivation positively affects learning satisfaction for face-to-face learning mode.

H4b. Motivation positively affects learning satisfaction for online learning mode.

H4c. Motivation positively affects learning satisfaction for blended learning mode.

3.4. Student performance

Du and Wu (2013) examined whether human interaction in blended learning enhances student performance and satisfaction for an introductory principles accounting course over the period 2010–201. It has been claimed that human interactions are a critical component of educational experience. Results indicate that human interactions do not impact student performance after controlling dimensions like transfer status, gender, GPA, and math grade. On the contrary, they found out that student performance improves with student interaction, suggesting that it is through learning satisfaction that a student's performance gets enhanced. Rakic, Tasic, & Marjanovic (2020) explore the usage and effect of virtual technologies on an online learning platform. They undertook their research with the Moodle learning management system. Data from the online learning platform were assessed empirically to pinpoint critical determinants of learners' performance in multiple courses. The findings indicate a significant relationship between the students' satisfaction and performance levels with the utilisation of digital academic resources from the e-learning platform.

Greller, Santally, Boojhawn, Rajablee, and Sungkur (2017) studied cases of online and blended learning at the University of Mauritius and ran the analysis of student interaction data against their achievements and performances. The sample entailed first-year students over two respective academic years of an undergraduate programme in multimedia and web velopment. The findings revealed that the final grades for the three online modules do not differ significantly, while there is a significant difference between the two on-campus modules. Analysing their engagement data, the findings (pass or fail) explicitly showed that the success rate increased, and, as a result, the failure rate went down as the regularity of students went up. The linear correlation coefficient between the different types of modes of assessment and the engagement degree is moderate and positive. Hence, they concluded that regularity featured both in face-to-face and digital participation influences considerably the overall SP. The students in the face-to-face courses, as contrasted against the virtual modules, had normalised total marks that were not significantly different on average. Along the same lines,

Sockalingam (2013) reports the interrelatedness between learners' satisfaction with the five dimensions of blended learning and their performance. The five dimensions have been: satisfaction with course design, online discussion, course content, assessment, and course handbook. The sample consisted of 2719 students from twelve business subjects taught in blended format at a Singapore university. The participants replied to a 15-item questionnaire by the end of their subject. Findings reveal that the causal model validated satisfaction with course design, content, and guide as significant input elements, whereas satisfaction with virtual discussion and evaluation methods as process factors and student grade as output performance elements. Out of the two process factors, assessment satisfaction had a direct positive impact on the subject marks but only explained 1.3% of the grade variance. Yet, assessment satisfaction was directly and positively influenced by curriculum content, online discussion, and course design.

Michelle (2011) used the Noel-Levitz Student Satisfaction Inventory (SSI) to evaluate traditional learning satisfaction in 12 various areas pertinent to brick-and-mortar academic entities (e.g., admissions, student services). She examined whether student satisfaction in these areas was relevant to persistence or student grades. Students with lower grades were reported to be less satisfied in multiple areas that they regarded as important: registration effectiveness, instructional effectiveness, student-centeredness, academic advising

and counselling, service excellence, concern for the individual, and campus climate. In traditional learning settings, student performance statistically related to satisfaction (and weight) ratings in multiple institutional areas. Hence, we hypothesize:

H5a. Learning satisfaction positively affects student performance for face-to-face learning mode.

H5b. Learning satisfaction positively affects student performance for online learning mode.

H5c. Learning satisfaction positively affects student performance for blended learning mode.

4. Methodology

4.1. Population and sample

For this study, we chose the quantitative approach and targeted business administration students (BA) at Omani universities to fill out a questionnaire for data collection. Regarding the pre-test, after formulating the questionnaire, the researchers of the present study sought the input of seven scholarly experts to evaluate the survey. The researchers specifically requested these experts to assess the comprehensibility and lucidity of the questions, as well as the soundness of the items, scales, and content. The format of the questionnaire was modified, and the explanations in each section were somewhat adjusted based on the feedback received. However, we did not make any modifications to the scales used in the questionnaire. The modifications mostly centre around incorporating comprehensive descriptions, which serve to introduce each segment and elucidate the objectives of the questions. Following this, a pilot study was undertaken in order to assess the face validity and minimise measurement error, as well as to discover any unclear eligibility requirements and misunderstandings regarding the survey questions. In accordance with the guidelines proposed by Nunnally (1994), a preliminary survey was administered to a sample of 20 students one week prior to the planned commencement of the primary data-gathering phase. The survey items underwent slight modifications based on the findings obtained from the analysis. The current study used longitudinal data from three survey copies that were distributed to the same student to compare face-to-face, online, and blended modes over three different time periods (Spring 2019, Spring 2020; Spring 2021). The population of the study came from BA students at Omani universities, and their enrollment was completely random. According to Abou-Shouk, Mannaa, and Elbaz (2021), experts feel that university students should have a decent level of knowledge that is sufficient to enable them to distinguish between the three different learning styles (i.e., face-to-face, online, and blended). In addition to this, accessing volunteers from inside this group is a simple and uncomplicated process (i.e., university students). The researchers asked for assistance in randomising the distribution of the questionnaire among the students, so they hired two research assistants to accomplish it for them. Thus, the current study utilised this method of simple random sampling. This is a common type of probability sampling that helps reduce bias in participant selection and improves the representation of the target population. It is also suitable for the collection of data from large geographical areas via electronic mediums (like an online survey), making it ideal for large-scale research projects (Hair, Sarstedt, Ringle, & Gudergan, 2017).

4.2. Data collection

BA students were emailed a link to gather data face-to-face and via an online survey. By May 2022, this technique for data collection had been completed. In accordance with the guidelines of Kock and Hadaya (2018) and Kock (2022), the WarpPLS statistical power and sample size requirement's function showed that the required sample size should be at least 160. The data acquired in this survey equals the criterion of 160, with a total of 160 replies received from business university students across Oman. Table 1 outlines the characteristics of the sample used in this research.

Table 1
Sample characteristics.

Characteristic	Face-to-face		Online		Blended	
	(n = 160) %		(n = 160) %		(n = 160) %	
Year	2019		2020		2021	
Gender						
Male	59	36.9	59	36.9	59	36.9
Female	101	63.1	101	63.1	101	63.1
Academic level						
Foundation year	35	21.9	35	21.9	35	21.9
1st and 2nd year	56	35.0	56	35.0	56	35.0
3rd and 4th year	69	43.1	69	43.1	69	43.1
GPA						
Fail	13	8.1	11	6.9	25	15.6
Pass	16	10.0	32	20.0	17	10.6
Good	49	30.6	57	35.6	43	26.9
Very good	47	29.4	40	25.0	41	25.6
Excellent	35	21.9	20	12.5	34	21.3

4.3. Measures

Ease of use, perceived usefulness, and learning motivation were derived from Hwang and Chang (2021) and Hwang et al. (2013a, 2013b), whereas learning satisfaction was constructed by Hwang and Chang (2021) and Sun, Tsai, Finger, Chen, and Yeh (2008). In addition, student academic performance was measured by the grade point average earned within the same academic subject. Students take exams in all classes and their performance is evaluated uniformly across courses owing to well-stated grading criteria. The following scoring breakdown (with corresponding weights) applies to all courses in this manner: Individual evaluations, attendance, and punctuality: 10%; class participation and the final examination: 50%; the midterm examination: 20%; group projects, article or case presentations, and exercises: 20%; and the total: 100%. (D'Alessio et al., 2019).

5. Data analysis and results

This study's analysis utilised the nonlinear partial least squares structural equation simulation (PLS-SEM) with WarpPLS version 8.0 software. This method is chosen because it permits the development of complicated conceptual frameworks for multi-block analysis, as argued by Kock (2022). Tables 1 and 2, respectively, describe the quality of the outer and inner models.

5.1. The measurement model

Prior to delving into the measurement model, the discriminant and convergent validity of each construct, in addition to their reliabilities (Cronbach's alpha) and composite reliabilities (CR), were all examined and analysed thoroughly. Because of this, we conducted an analysis of the factor loadings, as well as the CR and values, the average variance extracted (AVE), and the variance

Table 2
Internal consistency of reflective constructs across contexts.

Construct/Items	Face-to-face mode				Online mode				Blended mode			
	Loading	Mean	SD	CR/Ca/AVE/VIF	Loading	Mean	SD	CR/Ca/AVE/VIF	Loading	Mean	SD	CR/Ca/AVE/VIF
Perceived usefulness (PU)	**(0.848)**	**2.9**	**1.0**	**0.939/0.922/0.721/3.6**	**(0.777)**	**2.8**	**0.9**	**0.929/0.869/0.606/3.5**	**(0.868)**	**3.0**	**1.0**	**0.949/0.935/0.755/4.6**
PU1	(0.842)	2.9	1.0		(0.797)	2.8	1.2		(0.862)	3.0	1.4	
PU2	(0.892)	2.9	1.1		(0.775)	2.8	1.2		(0.855)	3.1	1.2	
PU3	(0.843)	3.0	0.9		(0.830)	2.7	1.1		(0.862)	3.0	1.3	
PU4	(0.805)	2.9	0.9		(0.813)	2.8	1.2		(0.876)	3.0	1.3	
PU5	(0.873)	3.0	1.0		(0.744)	2.7	1.3		(0.872)	3.1	1.3	
PU6	(0.835)	2.8	1.1		(0.703)	2.7	1.1		(0.884)	2.9	1.3	
Perceived ease of use (PEU)	**(0.815)**	**2.8**	**1.0**	**0.923/0.899/0.665/4.1**	**(0.789)**	**2.8**	**0.9**	**0.909/0.879/0.625/4.5**	**(0.855)**	**2.9**	**1.0**	**0.942/0.926/0.732/4.9**
PEU1	(0.787)	2.7	1.2		(0.829)	2.8	1.2		(0.865)	2.8	1.3	
PEU2	(0.800)	2.9	1.1		(0.802)	2.8	1.2		(0.838)	2.9	1.1	
PEU3	(0.834)	2.9	1.2		(0.797)	2.9	1.2		(0.873)	3.0	1.3	
PEU4	(0.826)	2.8	1.2		(0.850)	2.8	1.2		(0.844)	3.0	1.2	
PEU5	(0.834)	2.9	1.2		(0.738)	2.6	1.2		(0.883)	3.0	1.3	
PEU6	(0.811)	2.9	1.2		(0.721)	2.8	1.1		(0.827)	3.1	1.2	
Learning motivation (LMO)	**(0.802)**	**2.8**	**1.0**	**0.936/0.921/0.645/3.8**	**(0.805)**	**2.7**	**0.9**	**0.937/0.923/0.650/4.1**	**(0.839)**	**2.9**	**1.0**	**0.950/0.940/0.706/3.8**
LMO1	(0.765)	2.9	1.2		(0.759)	2.8	1.1		(0.837)	2.7	1.3	
LMO2	(0.792)	2.8	1.1		(0.808)	2.8	1.2		(0.817)	3.0	1.1	
LMO3	(0.794)	2.9	1.0		(0.849)	2.7	1.2		(0.837)	3.0	1.1	
LMO4	(0.826)	2.8	1.0		(0.817)	2.9	1.1		(0.830)	2.9	1.2	
LMO5	(0.827)	2.9	1.0		(0.857)	2.8	1.2		(0.845)	3.0	1.2	
LMO6	(0.799)	2.9	9.0		(0.807)	2.8	1.2		(0.849)	3.0	1.3	
LMO7	(0.834)	2.8	1.0		(0.775)	2.6	1.2		(0.838)	3.0	1.3	
LMO8	(0.784)	3.0	1.0		(0.771)	2.6	1.1	**0.928/**	(0.865)	3.1	1.2	
Learning satisfaction (LSA)	**(0.808)**	**2.8**	**0.9**	**0916/0.890/0.645/3.4**	**(0.825)**	**2.8**	**0.9**	**0.907/0.685/3.9**	**(0.854)**	**2.9**	**1.0**	**0.942/0.926/0.731/3.8**
LSA1	(0.802)	2.7	1.0		(0.736)	2.8	1.2		(0.852)	2.7	1.3	
LSA2	(0.818)	2.8	0.9		(0.738)	2.8	1.1		(0.859)	2.8	1.2	
LSA3	(0.800)	3.0	1.0		(0.868)	2.9	1.2		(0.871)	2.9	1.2	
LSA4	(0.833)	2.9	0.9		(0.871)	2.7	1.2		(0.854)	3.0	1.2	
LSA5	(0.799)	2.9	0.9		(0.868)	2.9	1.2		(0.854)	2.9	1.3	
LSA6	(0.801)	2.8	1.0		(0.871)	2.7	1.2		(0.838)	2.9	1.3	

CR = Composite reliability; Ca = Cronbach's alpha; AVE = average variance extracted; VIF = variance inflation factor.

inflation factor (VIF). As a consequence of this, we conducted an analysis of the factor loadings, as well as the CR and values, the average variance extracted (AVE), and the variance inflation factor (VIF). According to Table 2, all three measures of reliability and validity (CR, AVE, and α) were able to achieve a level of success that was higher than the required criteria of 0.7 and 0.5, respectively (Hair et al., 2017). In addition, multicollinearity and common method bias were investigated using Harman's single factor test and VIF scores (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003). Because the VIF was lower than five for every instrument, it appears that there was no major collinearity. Harman's single factor also produced a respectable result, demonstrating that no one factor explained more than fifty percent of the variation in the data. According to Hair et al. (2017), the heterotrait-monotrait (HTMT) correlation ratio was utilised to test discriminant validity. This ratio is becoming more widely regarded as a reliable method. As a result, each of the numbers presented in Table 3 falls below the HTMT threshold of 0.90. (Kock, 2022). In addition, the square roots of the AVE were examined to determine discriminant validity. Hence, the HTMT and AVEs were good, indicating adequate discriminant validity.

5.2. The structural model

In the inner model, the p-values and path coefficients (β) of the expected associations are introduced (see Fig. 1). In this vein, the structural findings for the three learning models—face-to-face, online, and blended—will be presented in the sections that follow (see Fig. 2 and Table 4).

5.2.1. Face-to-face learning

For the face-to-face model, the findings indicate that students perceived ease of use has a strong positive effect on their perceived usefulness ($\beta1 = 0.845$ and $p < 0.01$) and motivation ($\beta1 = 0.470$ and $p < 0.01$), signifying that H1 and H2 are accepted. Similarly, students perceived usefulness positively affects their motivation ($\beta1 = 0.369$ and $p < 0.01$). Therefore, we accept H3. Motivation in turn significantly influences students learning satisfaction ($\beta1 = 0.843$ and $p < 0.01$), indicating the acceptance of H4. However, students learning satisfaction slightly enhances their' academic performance ($\beta1 = 0.163$ and $p < 0.05$). Therefore, H5 is supported. Generally, the perceptions of students' ease of use and perceived usefulness explain 65% of the variance in students' motivation to use face-to-face learning ($R^21 = 0.65$), while students' motivation to use face-to-face learning explains 71% of the variance in their learning satisfaction ($R^21 = 0.71$). However, students learning satisfaction only explains 3% of the variance in students' academic performance ($R^21 = 0.03$).

5.2.2. Online learning

The findings of the online learning model reveal that students perceived ease of use has a strong positive effect on their perceived usefulness ($\beta2 = 0.858$ and $p < 0.01$), and motivation ($\beta2 = 0.462$ and $p < 0.01$), signifying that H1 and H2 are accepted. Students perceived usefulness positively affects their motivation ($\beta2 = 0.400$ and $p < 0.01$). Hence H3 is accepted. Motivation in turn significantly influences students learning satisfaction ($\beta2 = 0.844$ and $p < 0.01$), indicating the acceptance of H4. Accordingly, students learning satisfaction strongly enhances students' academic performance ($\beta2 = 0.562$ and $p < 0.01$), signifying, the support of H5. Generally, the perceptions of student's ease of use and perceived usefulness explain 69% of the variance in student's motivation to use online learning ($R^22 = 0.65$), while student's motivation to use online learning explains 71% of the variance in their learning satisfaction ($R^22 = 0.71$). Thus, students learning satisfaction strongly explains 32% of the variance in students' academic performance ($R^22 = 0.32$).

5.2.3. Blended learning

The outcomes of the blended learning model indicate that students perceived ease of use has a strong positive effect on their perceived usefulness ($\beta3 = 0.858$ and $p < 0.01$), and motivation ($\beta3 = 0.551$ and $p < 0.01$), signifying that H1 and H2 are accepted. Students perceived usefulness positively affects their motivation ($\beta3 = 0.372$ and $p < 0.01$). Therefore, H3 is accepted. Motivation in turn significantly influences students learning satisfaction ($\beta3 = 0.846$ and $p < 0.01$), indicating the acceptance of H4. Unexpectedly,

Table 3
Assessment of Discriminant Validity using HTMT.

Data set	Constructs	1	2	3	4
Face to face	1. Perceived usefulness				
	2. Perceived ease of use	0.803			
	3. Learning motivation	0.821	0.853		
	4. Learning satisfaction	0.812	0.888	0.824	–
Online	1. Perceived usefulness				
	2. Perceived ease of use	0.880			
	3. Learning motivation	0.888	0.895		–
Blended	4. Learning satisfaction	0.787	0.812	0.828	
	1. Perceived usefulness				
	2. Perceived ease of use	0.814			
	3. Learning motivation	0.898	0.830		
	4. Learning satisfaction	0.837	0.821	0.805	

HTMT ratios (good if < 0.90, best if < 0.85).

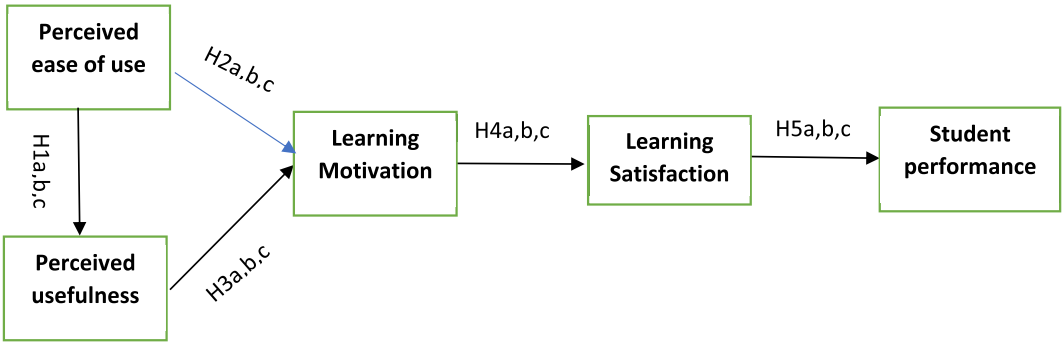

Fig. 1. Conceptual framework
a "Face-to-face learning mode"
b "Online learning mode''
c "Blended learning mode''

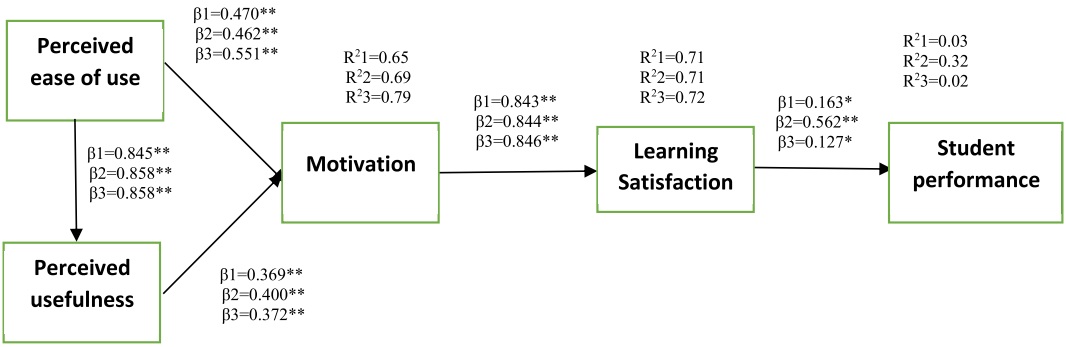

Fig. 2. Structural model
**Significance level at 99% (*p-value* < 0.01).
*Significance level at 95% (*p-value* < 0.05).

Table 4
Structural model assessment results.

No.	Hypothesis	Beta	*P-Value*	R^2	f^2	Decision
H1a	Perceived ease of use positively affects the perceived usefulness of face-to-face learning mode.	0.470[a]	<0.001	0.68	0.714	Supported
H1b	Perceived ease of use positively affects the perceived usefulness of online learning mode.	0.462[a]	<0.001	0.73	0.737	Supported
H1c	Perceived ease of use positively affects the perceived usefulness of the blended learning mode.	0.551[a]	<0.001	**0.73**	0.736	Supported
H2a	Perceived ease of use positively affects perceived motivation face-to-face learning mode.	0.470[a]	<0.001	0.65	0.366	Supported
H2b	Perceived ease of use positively affects perceived motivation for online learning mode.	0.462[a]	<0.001	0.69	0.371	Supported
H2c	Perceived ease of use positively affects perceived motivation for blended learning mode.	0.551[a]	<0.001	0.79	0.478	Supported
H3a	Perceived usefulness positively affects motivation for face-to-face learning mode.	0.369[a]	<0.001	0.65	0.280	Supported
H3b	Perceived usefulness positively affects motivation for online learning mode.	0.400[a]	<0.001	0.69	0.318	Supported
H3c	Perceived usefulness positively affects motivation for blended learning mode.	0.372[a]	<0.001	0.79	0.313	Supported
H4a	Motivation positively affects learning satisfaction for face-to-face learning mode.	0.843[a]	<0.001	0.71	0.710	Supported
H4b	Motivation positively affects learning satisfaction for online learning mode.	0.844[a]	<0.001	0.71	0.713	Supported
H4c	Motivation positively affects learning satisfaction for blended learning mode.	0.846[a]	<0.001	0.72	0.715	Supported
H5a	Learning satisfaction positively affects student performance for face-to-face learning mode.	0.163[b]	0.017	0.03	0.027	Supported
H5b	Learning satisfaction positively affects student performance for online learning mode.	0.562[a]	<0.001	0.32	0.315	Supported
H5c	Learning satisfaction positively affects student performance for blended learning mode.	0.127[b]	= 0.050	0.02	0.016	Supported

[a] Significance level at 99% (*p-value* < 0.01).
[b] Significance level at 95% (*p-value* < 0.05).

students learning satisfaction slightly enhances students' academic performance ($\beta 3 = 0.127$ and $p < 0.05$), signifying, the support of H5. Generally, the perceptions of students' ease of use and perceived usefulness explain 79% of the variance in students' motivation to use online learning ($R^2 3 = 0.79$). Students' motivation to use blended learning explains 72% of the variance in their learning satisfaction ($R^2 3 = 0.72$). However, students learning satisfaction weakly explains 02% of the variance in students' academic performance ($R^2 3 = 0.02$).

5.2.4. Indirect and direct effects

For the face-to-face delivery mode, we calculated the direct and indirect effects using the bootstrapping method in WarpPLS with n = 999 bootstrap resamples and a 95% confidence interval (Kock, 2022). As for the direct paths, the results indicated that perceived ease of use on student performance is significant (β = 0.149) (see Table 5). On the other hand, perceived usefulness has no significant influence on student performance (β = −0.018). Also, learning motivation has no significant influence on student performance (β = −0.118). As for the indirect paths, the results showed that perceived ease of use on student performance is not significant (β = 0.065). On the other hand, perceived usefulness has no significant influence on student performance (β = 0.015). The results showed that learning motivation has a significant effect on student performance (=-0.118).

The study calculated the direct and indirect effects of the online delivery mode (see Table 5). As for the direct paths, the results implied that perceived ease of use on student performance is not significant (β = 0.028). On the other hand, perceived usefulness has a significant influence on student performance (β = 0.450). Also, learning motivation has a significant influence on student performance (β = −0.400). As for the indirect paths, the results showed that perceived ease of use on student performance is significant (β = 0.219). Also, perceived usefulness has a significant influence on student performance (β = 0.190). The results also indicated that learning motivation has a significant effect on student performance (β = 0.474).

Furthermore, the study calculated the direct and indirect effects of the blended delivery mode (see Table 5). As for the direct paths, the results implied that perceived ease of use, usefulness, and learning motivation on student performance are not significant (β = −0.059, β = 0.012, and β = 0.097 subsequently). As for the indirect paths, the results showed that perceived ease of use on student performance is not significant (β = 0.059). Also, perceived usefulness has no significant influence on student performance (β = 0.040). On the other hand, the results showed that learning motivation has a significant impact on student performance (=0.107).

6. Discussion

In addition to face-to-face learning, two major forms of educational delivery techniques have grown in popularity in recent years, namely online and blended learning. Each form of learning has certain ramifications, benefits, and drawbacks of its own. In this paper, we developed and tested a conceptual model to compare the factors that determine student performance in the Omani higher education context through these three educational delivery techniques. In particular, these types were investigated based on the perspectives of perceived ease of use, perceived usefulness, learning motivation, and student satisfaction. Surprisingly, though these findings have both theoretical and practical implications for higher education institutions.

6.1. Theoretical implications

This work offers three significant theoretical contributions. The authors demonstrate that perceived ease of use and perceived usefulness significantly drive students' motivation to use different learning modes (face-to-face, online, and blended) in higher education institutions, building on previous research on learning mode impact and its antecedents. This supports the favourable impact of perceived ease of use and perceived usefulness among the three modes (Lin et al., 2010). Further, the present study's findings make a valuable contribution to the existing body of literature on students' satisfaction in terms of learning modes. Specifically, our research

Table 5
Indirect and direct effects.

Indirect Path	Indirect effect	P value	Direct Path	Direct path	P value
Face-to-face delivery mode			**Face-to-face delivery mode**		
PEOU — > PU, LM & LS— > SP	0.065	0.071	PEOU — > SP	0.149	0.027[b]
PU — > LM & LS— > SP	0.051	0.132	PU — > SP	−0.018	0.411
PEOU — > PU— > LM	0.311	<0.001[a]			
PU — > LM — > LS	0.311	<0.001[a]			
PEOU — > PU — > LS	0.396	<0.001[a]			
LM — > LS — > SP	0.137	.006[a]	LM — > SP	−0.108	0.082
Online delivery mode			**Online delivery mode**		
PEOU — > PU, LM & LS— > SP	0.219	<0.001[a]	PEOU — > SP	0.028	0.360
PU — > LM & LS— > SP	0.190	<0.001[a]	PU — > SP	0.650	<0.001[a]
PEOU — > PU— > LM	0.338	<0.001[a]			
PU — > LM — > LS	0.343	<0.001[a]			
PEOU — > PU — > LS	0.390	<0.001[a]			
LM — > LS — > SP	0.474	<0.001[a]	LM — > SP	0.400	<0.001[a]
Blended delivery mode			**Blended delivery mode**		
PEOU — > PU, LM & LS— > SP	0.059	0.096	PEOU — > SP	−0.059	0.225
PU — > LM & LS— > SP	0.040	0.189	PU — > SP	0.012	0.440
PEOU — > PU— > LM	0.315	<0.001[a]			
PU — > LM — > LS	0.319	<0.001[a]			
PEOU — > PU — > LS	0.466	<0.001[a]	LM — > SP	0.097	0.105
LM — > LS — > SP	0.107	0.026[a]			

[a] Significance level at 99% (*p-value* < 0.01).
[b] Significance level at 95% (*p-value* < 0.05).

demonstrates that students' desire to engage in face-to-face learning significantly influences their satisfaction levels, yet it is important to note that this influence has only a little effect on their academic achievement. Nevertheless, the motivation of students to utilize online learning platforms in blended learning modes has a notable effect on their level of satisfaction. Nevertheless, it was observed that students in blended learning had comparatively lower levels of academic achievement when compared to students in alternative modes of learning. The possible disparities in instructional approaches and modalities between the online and offline components of the course may contribute to incongruous educational achievements among learners (Banihashem, Noroozi, den Brok, Biemans, & Kerman, 2023).

Our work provides empirical evidence linking various variables to student achievement in a variety of classroom settings where such a correlation has not previously been shown, thus contributing to the literature. This method allows for an in-depth comprehension of how the various forms of instruction really affect students' academic outcomes. This further highlights the need to understand the unique philosophical foundations of each of these methods. While our study expands on other studies that have highlighted the uniqueness of these methods (Müller & Mildenberger, 2021; Tarhini, Hone, Liu, & Tarhini, 2017), the Omani environment we've chosen to explore gives our research a new perspective for studies in the same region. This emphasises the importance of understanding the theoretical implications of these varied pedagogical approaches if we need to create effective and interesting learning environments for students (Müller & Mildenberger, 2021).

6.2. Practical implications

Using a variety of learning methods could generally encourage interactive learning, which entails getting students involved in practical, group, and problem-based education systems. This may increase learners' motivation and satisfaction, as well as student engagement and performance. For instance, technologies can be included in instructional and learning procedures as part of educational delivery methods. By doing so, teaching may become more exciting and participatory with the use of integrated multimedia resources, including gaming, simulation-based learning, and webinars. In this vein, learners can pursue online education at their own pace and on their own schedules using blended learning, giving them more flexibility. However, the absence of face-to-face interaction may result in less direct interaction between instructors and learners, which may influence the standard of teaching. Therefore, digital communication methods like videoconferences can help with this problem. In addition, to use these technologies successfully, different initiatives must be involved, such as well-planned learning management systems, digital resources, and effective multimedia technologies.

Education providers should note that it may be necessary for academic staff to acquire novel expertise and skills competencies in order to have a successful implementation of any new educational delivery methods. To assist instructors in this endeavour, educational organisations might offer chances for career development. Furthermore, it is crucial for universities to provide efficient procedures for determining the effect of educational delivery systems. This could involve assessing student academic performance and pinpointing areas that require enhancement using data analytics, user polling, and other methods of feedback.

Overall, adopting the appropriate educational delivery method can provide universities and colleges with various benefits, such as increased flexibility, accessibility, and student engagement. In order to ensure the best possible learning outcomes for students, rigorous and effective planning, teacher preparation, and regular assessments and evaluations are also essential.

6.3. Limitations, future studies, and conclusion

The study's sample, which is restricted to a few institutions in Oman, needs to be expanded to include populations with a wider range of socioeconomic, educational, demographic, and psychological characteristics. This would increase the generalizability of the study's findings. Future research may thus broaden the sample to include students from other towns and possibly other nations. Finally, only local students were the subject of our investigation. Future research may evaluate the views of local students in comparison to those of foreign students who were unable to attend their study face-to-face during the COVID-19 outbreak.

In conclusion, the partial least squares method (PLS-SEM) was used to investigate students' academic performance across blended learning, online learning, and face-to-face learning based on four dimensions: perceived usefulness, perceived ease of use, learning motivation, and students' satisfaction. The study concluded that each type of learning—online, face-to-face, and blended—has specific benefits and implications. Decision-makers can select the most effective delivery strategy for their students and establish a productive learning environment by being aware of the advantages and disadvantages of each form of learning.

CRediT authorship contribution statement

Islam Elbayoumi Salem: Conceptualization, Formal analysis, Methodology, Validation, Writing – review & editing. **Alamir AL-Alawi:** Conceptualization, Data curation, Project administration, Supervision. **Sanyo Moosa:** Conceptualization, Investigation, Project administration, Supervision. **Lujain El-Maghraby:** Conceptualization, Resources, Writing – original draft, Writing – review & editing. **Nasser Alhamr Alkathiri:** Conceptualization, Data curation, Investigation, Resources, Validation, Visualization. **Ahmed Mohamed Elbaz:** Conceptualization, Formal analysis, Investigation, Software, Validation, Visualization, Writing – review & editing.

Declarations of competing interest

None.

Data availability

Data will be made available on request.

Acknowledgements

The authors would like to express their gratitude to the University of Technology and Applied Sciences, Salalah, Sultanate of Oman. This research project was funded by the University of Technology and Applied Sciences through the internal Research funding program, grant No 05-IRFP-2022.

APPENDIX A

The reality of Face-to-face learning on student performance in Omani universities

Dear Student …

This questionnaire provides the basis for the learning mode (face-to-face) that plays an important role in student performance in Omani universities. The aim of this study is to investigate how the mode of learning (face-to-face) influences student performance before, during, and after COVID-19 in Omani universities.

This study is purely academic and the data you provide will be used only for scientific research and will be kept strictly confidential. It will help in gaining a better understanding **of** the influence of learning modes on student performance. An executive summary of the research major findings can be sent via email. Your kind cooperation in this research is very much appreciated and the research team sincerely hopes that you will find the study of interest to you.

We deeply appreciate your valuable time and cooperation.

First part: Respondents' profile & Demographic data

Gander	(1)- female	(2)- male	
Academic level	(1) foundation years	(2) first -second years	(3) third- four year
Major	(1)- business administration	(2)-mass communication	(3)- information technology
SGPA SP 2019: (1) 4.00–3.50 (2) 3.49–3.00 (3) 2.99–2.50 (4) 2.49–2.00 (5) less than 2.00			

Second Part: face to face learning and student performance

Please assess to what extent the following statements related to the influence of study modes on student performance. Indicate the degree of agreement or disagreement that fits the situation in your college.

Please circle one choice for each of the following statements (1 = Strongly disagree … 5 = strongly agree).

The reality of blended learning		SD	Dis	Neu	Ag	SA
		A. 1	2	3	4	5
1. Face-to-face learning and performance						
1.1 Perceived Usefulness						
PU1	This way of learning in class enriches learning activities.	1	2	3	4	5
PU2	This way of learning in class is very helpful for me to acquire new knowledge.	1	2	3	4	5
PU3	The learning mechanism provided by this way of learning in class makes the learning process smoother.	1	2	3	4	5
PU4	This way of learning in class helps me get useful information when I need it.	1	2	3	4	5
PU5	This way of learning in class helps me learn better.	1	2	3	4	5
PU6	This way of learning in class is more useful than the traditional computer classroom.	1	2	3	4	5
1.2 Perceived Ease of Use						
PEOU1	The kind of operating system by this way of learning in class is not difficult for me.	1	2	3	4	5
PEOU2	It only took me a short time to fully understand how to apply this way of learning in class.	1	2	3	4	5
PEOU3	The learning activities in this way of learning in class are easy to understand and follow.	1	2	3	4	5
PEOU4	I quickly learned to apply this way of learning in class	1	2	3	4	5
PEOU5	It is not difficult for me to apply the learning system in this way of learning in class.	1	2	3	4	5
PEOU6	I think the system interface of this way of learning in class is easy to use.	1	2	3	4	5
1.3 Learning Motivation						
LM1	I think this way of learning in class is interesting	1	2	3	4	5
LM2	Employees are valued for their individual expertise.	1	2	3	4	5
LM3	I want to learn more in this way of learning in class.	1	2	3	4	5

(*continued on next page*)

(*continued*)

The reality of blended learning		SD	Dis	Neu	Ag	SA
		A. 1	2	3	4	5
LM4	I think it is worth applying this way of learning in class.	1	2	3	4	5
LM5	To me, it is important to apply this way of learning in class.	1	2	3	4	5
LM6	I know that learning to apply this way of learning in class is very important in the future.	1	2	3	4	5
LM7	I will seek more information to learn how to apply this way of learning in class.	1	2	3	4	5
LM8	I think it is important for every student to learn to apply this way of learning in class.	1	2	3	4	5
1.4 Learning Satisfaction						
LS1	I am satisfied with this way of learning in class.	1	2	3	4	5
LS2	If I still have the opportunity to apply this way of learning in class, I will be happy to do so.	1	2	3	4	5
LS3	I think it is a wise choice to study courses in this way of learning in class.	1	2	3	4	5
LS4	I feel very satisfied with this way of learning in class.	1	2	3	4	5
LS5	I think this way of learning in class satisfies my learning needs very well.	1	2	3	4	5
LS6	I will try to apply this way of learning in class as much as possible to study courses.	1	2	3	4	5

Any other suggestions, please add.

The reality of online learning on student performance in Omani universities. learning on student performance in Omani universities

Dear Student …

This questionnaire provides the basis for the learning mode (Online) that plays an important role in student performance in Omani universities. The aim of this study is to investigate how the mode of learning (Online) influences student performance before, during, and after COVID-19 in Omani universities.

This study is purely academic and the data you provide will be used only for scientific research and will be kept strictly confidential. It will help in gaining a better understanding **of** the influence of learning modes on student performance. An executive summary of the research major findings can be sent via email. Your kind cooperation in this research is very much appreciated and the research team sincerely hopes that you will find the study of interest to you.

We deeply *appreciate your valuable time and cooperation.*

First part: Respondents' profile & Demographic data

Gander	(1)- female	(2)- male	
Academic level	(1) foundation years	(2) first -second years	(3) third- four year
Major	(1)- business administration	(2)-mass communication	(3)- information technology
SGPA SP 2020: (1) 4.00–3.50 (2) 3.49–3.00 (3) 2.99–2.50 (4) 2.49–2.00 (5) less than 2.00			

Second Part: Online learning and student performance

Please assess to what extent the following statements related to the influence of study modes on student performance. Indicate the degree of agreement or disagreement that fits the situation in your college.

Please circle one choice for each of the following statements (1 = Strongly disagree … 5 = strongly agree).

Reality of blended learning		SD	Dis	Neu	Ag	SA
		B. 1	2	3	4	5
1. Online learning and performance						
1.1 Perceived Usefulness						
PU1	This way of learning in online enriches learning activities.	1	2	3	4	5
PU2	This way of learning in online is very helpful for me to acquire new knowledge.	1	2	3	4	5
PU3	The learning mechanism provided by this way of learning in online makes the learning process smoother.	1	2	3	4	5
PU4	This way of learning in online helps me get useful information when I need it.	1	2	3	4	5
PU5	This way of learning in online helps me learn better.	1	2	3	4	5
PU6	This way of learning in online is more useful than the traditional computer classroom.	1	2	3	4	5
1.2 Perceived Ease of Use						
PEOU1	The kind of operating system by this way of learning in online is not difficult for me.	1	2	3	4	5
PEOU2	It only took me a short time to fully understand how to apply this way of learning in online	1	2	3	4	5
PEOU3	The learning activities in this way of learning in online are easy to understand and follow.	1	2	3	4	5
PEOU4	I quickly learned to apply this way of learning in online	1	2	3	4	5
PEOU5	It is not difficult for me to apply the learning system in this way of learning in online	1	2	3	4	5
PEOU6	I think the system interface of this way of learning in online is easy to use.	1	2	3	4	5
1.3 Learning Motivation						
LM1	I think this way of learning in online is interesting	1	2	3	4	5

(*continued on next page*)

(*continued*)

Reality of blended learning		SD	Dis	Neu	Ag	SA
		B. 1	2	3	4	5
LM2	Employees are valued for their individual expertise.	1	2	3	4	5
LM3	I want to learn more in this way of learning in online.	1	2	3	4	5
LM4	I think it is worth to apply this way of learning in online.	1	2	3	4	5
LM5	To me, it is important to apply this way of learning in online.	1	2	3	4	5
LM6	I know that learning to apply this way of learning in online is very important in the future.	1	2	3	4	5
LM7	I will seek more information to learn how to apply this way of learning in online	1	2	3	4	5
LM8	I think it is important for every student to learn to apply this way of learning in online	1	2	3	4	5
1.4 Learning Satisfaction						
LS1	I am satisfied with this way of learning in online.	1	2	3	4	5
LS2	If I still have the opportunity to apply this way of learning in online, I will be happy to do so.	1	2	3	4	5
LS3	I think it is a wise choice to study courses in this way of learning in online.	1	2	3	4	5
LS4	I feel very satisfied with this way of learning in online.	1	2	3	4	5
LS5	I think this way of learning in online satisfies my learning needs very well.	1	2	3	4	5
LS6	I will try to apply this way of learning in online as much as possible to study courses.	1	2	3	4	5

Any other suggestions, please add.

The reality of blended learning on student performance in Omani universities

Dear Student …

This questionnaire provides the basis for the learning mode (blended) that plays an important role in student performance in Omani universities. The aim of this study is to investigate how the mode of learning (blended) influences student performance before, during, and after COVID-19 in Omani universities.

This study is purely academic and the data you provide will be used only for scientific research and will be kept strictly confidential. It will help in gaining a better understanding **of** the influence of learning modes on student performance. An executive summary of the research major findings can be sent via email. Your kind cooperation in this research is very much appreciated and the research team sincerely hopes that you will find the study of interest to you.

We deeply appreciate your valuable time and cooperation.

First part: Respondents' profile & Demographic data

Gander	(1)- female	(2)- male	
Academic level	(1) foundation years	(2) first -second years	(3) third- four year
Major	(1)- business administration	(2)-mass communication	(3)- information technology
SGPA SP 2021: (1) 4.00–3.50 (2) 3.49–3.00 (3) 2.99–2.50 (4) 2.49–2.00 (5) less than 2.00			

Second Part: blended learning and student performance

Please assess to what extent the following statements related to the influence of study modes on student performance. Indicate the degree of agreement or disagreement that fits the situation in your college.

Please circle one choice for each of the following statements (1 = Strongly disagree … 5 = strongly agree).

Reality of blended learning		SD	Dis	Neu	Ag	SA
		C. 1	2	3	4	5
1. Blended learning and performance						
1.1 Perceived Usefulness						
PU1	This way of learning in blended enriches learning activities.	1	2	3	4	5
PU2	This way of learning in blended is very helpful for me to acquire new knowledge.	1	2	3	4	5
PU3	The learning mechanism provided by this way of learning in online makes the learning process smoother.	1	2	3	4	5
PU4	This way of learning in blended helps me get useful information when I need it.	1	2	3	4	5
PU5	This way of learning in blended helps me learn better.	1	2	3	4	5
PU6	This way of learning in blended is more useful than the traditional computer classroom.	1	2	3	4	5
1.2 Perceived Ease of Use						
PEOU1	The kind of operating system by this way of learning in blended is not difficult for me.	1	2	3	4	5
PEOU2	It only took me a short time to fully understand how to apply this way of learning in blended	1	2	3	4	5
PEOU3	The learning activities in this way of learning in blended are easy to understand and follow.	1	2	3	4	5
PEOU4	I quickly learned to apply this way of learning in blended	1	2	3	4	5
PEOU5	It is not difficult for me to apply the learning system in this way of learning in blended	1	2	3	4	5

(*continued on next page*)

(*continued*)

Reality of blended learning		SD	Dis	Neu	Ag	SA
		C. 1	2	3	4	5
PEOU6	I think the system interface of this way of learning in blended is easy to use.	1	2	3	4	5
1.3 Learning Motivation						
LM1	I think this way of learning in blended is interesting	1	2	3	4	5
LM2	Employees are valued for their individual expertise.	1	2	3	4	5
LM3	I want to learn more in this way of learning in blended.	1	2	3	4	5
LM4	I think it is worth to apply this way of learning in blended.	1	2	3	4	5
LM5	To me, it is important to apply this way of learning in blended.	1	2	3	4	5
LM6	I know that learning to apply this way of learning in blended is very important in the future.	1	2	3	4	5
LM7	I will seek more information to learn how to apply this way of learning in blended	1	2	3	4	5
LM8	I think it is important for every student to learn to apply this way of learning in blended.	1	2	3	4	5
1.4 Learning Satisfaction						
LS1	I am satisfied with this way of learning in blended.	1	2	3	4	5
LS2	If I still have the opportunity to apply this way of learning in online, I will be happy to do so.	1	2	3	4	5
LS3	I think it is a wise choice to study courses in this way of learning in blended.	1	2	3	4	5
LS4	I feel very satisfied with this way of learning in blended.	1	2	3	4	5
LS5	I think this way of learning in blended satisfies my learning needs very well.	1	2	3	4	5
LS6	I will try to apply this way of learning blended as much as possible to study courses.	1	2	3	4	5

Any other suggestions, please add.

References

Abou-Shouk, M. A., Mannaa, M. T., & Elbaz, A. M. (2021). Women's empowerment and tourism development: A cross-country study. *Tourism Management Perspectives, 37*, Article 100782.

Al-Abri, M. A. (2015). Sleep Deprivation and Depression: A bi-directional association. *Sultan Qaboos University Medical Journal, 15*(1), 4.

Al-Harthi, A. (2002). Geohazard assessment of sand dunes between Jeddah and Al-Lith, western Saudi Arabia. *Environmental Geology, 42*, 360–369.

Allen, I. E., Seaman, J., & Garrett, R. (2007). *). Blending in: The extent and promise of blended education in the United States*. Sloan Consortium. PO Box 1238, Newburyport, MA 01950.

Anthony, B., Kamaludin, A., & Romli, A. (2022). Blended learning adoption and implementation in higher education: A theoretical and systematic review. *Technology, Knowledge and Learning, 27*, 531–578. https://doi.org/10.1007/s10758-020-09477-z

Bachtiar, F. A., Rachmadi, A., & Pradana, F. (2014). Acceptance in the deployment of blended learning as a learning resource in information technology and computer science program. South Kuta. In *Asia-pacific conference on computer aided system engineering (APCASE)* (pp. 131–135). Indonesia: Brawijaya university. https://doi.org/10.1109/APCASE.2014.6924486, 2014.

Banihashem, S. K., Noroozi, O., den Brok, P., Biemans, H. J., & Kerman, N. T. (2023). Modeling teachers' and students' attitudes, emotions, and perceptions in blended education: Towards post-pandemic education. *International Journal of Management in Education, 21*(2), Article 100803.

Baporikar, N., & Shah, I. A. (2012). Quality of higher education in 21st century–A case of Oman. *Journal of Educational and Instructional Studies in the World, 2*(2), 9–18.

Bernard, R. M., Borokhovski, E., Schmid, R. F., Tamim, R. M., & Abrami, P. C. (2014). A meta-analysis of blended learning and technology use in higher education: From the general to the applied. *Journal of Computing in Higher Education, 26*(1), 87–122. https://doi.org/10.1007/s12528-013-9077-3

Bouguebs, R. (2019). Approaching a reading course via moodle-based blended learning: EFL learners' insights. *Modern Journal of Language Teaching Methods, 9*(11). https://doi.org/10.26655/mjltm.2019.11.1

Bullen, M., & Janes, D. P. (2007). *Making the transition to e-learning: Strategies and issues*. Hershey, PA: Information Science Publishing.

Chen, J. (2022). Effectiveness of blended learning to develop learner autonomy in a Chinese university translation course. *Education and Information Technologies, 27*. https://doi.org/10.1007/s10639-022-11125-1

Chen, C. H., Lee, W. C., Ng, K. S., & Ngeau, J. Y. (2022). *Factors affecting higher education students' behavioral intention toward hybrid learning in Malaysian private universities*. UTAR: Final Year Project.

Daud, N. M., Liting, N. F. D., Nasir, N. R. M., Hadini, N., & Bahri, N. A. S. (2022). Preferences for online learning platform types: Does it matter to students' academic performance? *International Journal of Academic Research in Business and Social Sciences, 12*(12), 605–625.

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly: Management Information Systems, 13*(3), 319.

Dhawan, S. (2020). Online learning: A panacea in the time of COVID-19 crisis. *Journal of Educational Technology Systems, 49*(1), 5–22.

Donn, G. A. R. I., & Issan, S. (2007). Higher education in transition: Gender and change in the Sultanate of Oman. *Scottish Educational Review, 39*(2), 173–185.

Driscoll, M. (2002). Blended learning: Let's get beyond the hype. *E-learning, 1*(4), 1–4.

Dwijuliani. (2021). *Journal of Physics: Conference Series*. https://doi.org/10.1088/1742-6596/1810/1/012072

Famularsih, S. (2020). Students' Experiences in Using Online Learning Applications Due to COVID-19 in English Classroom. *Studies in Learning and Teaching, 1*(2), 112–121.

Fang, J., Pechenkina, E., & Rayner, G. M. (2023). Undergraduate business students' learning experiences during the COVID-19 pandemic: Insights for remediation of future disruption. *International Journal of Management in Education, 21*(1), Article 100763.

Fuady, I., Sutarjo, M. A., & Ernawati, S. (2021). Analysis of students' perceptions of online learning media during the covid-19 pandemic (study of E-learning media: Zoom, Google Meet, Google classroom, and LMS). *Randwick International of Social Science (RISS) Journal, 2*(1), 51–56. https://doi.org/10.47175/rissj.v2i1.177

Gecer, A., & Dag. (2015). A blended learning experience. *Funda Educational Sciences: Theory and Practice, 12*(1), 3818–3823. https://doi.org/10.1016/j.sbspro.2015.01.1119

Hair, J. F., Jr., Sarstedt, M., Ringle, C. M., & Gudergan, S. P. (2017). *Advanced issues in partial least squares structural equation modeling*. Sage publications.

Hanafizadeh, P., & Shaikh, A. A. (2021). Developing doctoral students'/researchers' understanding of the journal peer-review process. *International Journal of Management in Education, 19*(2), 1–16.

Howard, G. S., & Maxwell, S. E. (1980). Correlation between student satisfaction and grades: A case of mistaken causation? *Journal of Educational Psychology, 72*(6), 810–820.

Huang, C. H. (2021). Using PLS-SEM model to explore the influencing factors of learning satisfaction in blended learning. *Education Sciences, 11*(5), 249.

Hwang, G. J., & Chang, C. Y. (2021). A review of opportunities and challenges of chatbots in education. *Interactive Learning Environments*, 1–14.

Hwang, G. J., Yang, L. H., & Wang, S. Y. (2013a). A concept map-embedded educational computer game for improving students' learning performance in natural science courses. *Computers & Education, 69*, 121–130.
Hwang, G. J., Yang, L. H., & Wang, S. Y. (2013b). A concept map-embedded educational computer game for improving students' learning performance in natural science courses. *Computers & Education, 69*, 121–130.
Karjaluoto, H., Shaikh, A. A., Saarijärvi, H., & Saraniemi, S. (2019). How perceived value drives the use of mobile financial services apps. *International Journal of Information Management, 47*, 252–261.
Kaur, M. (2013). Blended learning - its challenges and future. *Procedia - Social and Behavioral Sciences, 93*, 612–617. https://doi.org/10.1016/j.sbspro.2013.09.248
KK, S., & Maskari, A. A. (2019). Student engagement in blended learning instructional design: an analytical study. *Learning and Teaching in Higher Education: Gulf Perspectives, 15*(2), 61–79.
Kock, N. (2022). *WarpPLS UserManual: Version 8.0, ScriptWarp systems. Laredo, TX*.
Kock, N., & Hadaya, P. (2018). Minimum sample size estimation in PLS-SEM: The inverse square root and gamma-exponential methods. *Information Systems Journal, 28*(1), 227–261.
Kumar. (2021). Blended learning tools and practices: A comprehensive analysis,". *IEEE Access, 9*, 85151–85197. https://doi.org/10.1109/ACCESS.2021.3085844
Kwon, C., & Choi, E. (2022). Extroversion and subjective well-being among Korean university students: the moderating role of attachment. *Psychological Reports, 125*(2), 1068–1085.
Mahande, R. D., Darmawan, F. A., & Malago, J. D. (2021). Metacognitive skill assessment model through the blended learning management system in vocational education. https://doi.org/10.21831/jpv.v11i1.36912.
Mali, D., & Lim, H. (2021). How do students perceive face-to-face/blended learning as a result of the Covid-19 pandemic? *International Journal of Management in Education, 19*(3), Article 100552.
Maulida, D. S., Rahman, M. A., Handrianto, C., & Rasool, S. (2022). A review of the blended learning as the model in improving students paragraph writing skills. Abjadia. *International Journal of Education, 7*(1), 59–72.
Mueller, F. A., & Wulf, T. (2022). Blended learning environments and learning outcomes: The mediating role of flow experience. *International Journal of Management in Education, 20*(3), Article 100694.
Müller, C., & Mildenberger, T. (2021). Facilitating flexible learning by replacing classroom time with an online learning environment: A systematic review of blended learning in higher education. *Educational Research Review, 34*, Article 100394.
Nunnally, J. C. (1994). *Psychometric theory 3E*. Tata McGraw-hill education.
Oe, H., Takemoto, T., & Ridwan, M. (2020). Is gamification a magic tool?: Illusion, remedy, and future opportunities in enhancing learning outcomes during and beyond the COVID-19. *Budapest International Research and Critics in Linguistics and Education (BirLE) Journal, 3*(3), 1401–1414.
Oh, J. E., Chan, Y. K., & Kim, K. V. (2020). Social Media and E-Portfolios: Impacting Design Students' Motivation through Project-Based Learning. *IAFOR Journal of Education, 8*(3), 41–58.
Peimani, N., & Kamalipour, H. (2022). The future of design studio education: Student experience and perception of blended learning and teaching during the global pandemic. *Education Sciences, 12*(2), 140.
Podsakoff, P. M., MacKenzie, S. B., Lee, J. Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology, 88*(5), 879.
Prohorets, E., & Plekhanova, M. (2015). Interaction intensity levels in blended learning environment. *Procedia - Social and Behavioral Sciences, 174*, 3818–3823.
Qunfei, X., Abdullah, N., & Mustapha, S. M. (2020). The Influencing Factors and Incentive Measures Of College Students' Online English Learning Motivation. *Journal of Education and Social Sciences, 15*(2), 74–81.
Riswanto, A., & Aryani, S. (2017). Learning motivation and student achievement: Description analysis and relationships both COUNS-edu. *The International Journal of Counseling and Education, 2*, 42.
Rodríguez, J. M. M., & Rojo, A. S. (2020). *Blended learning: Convergence between technology and pedagogy*. Netherlands: Springer.
Shaikh, A. A., Glavee-Geo, R., Karjaluoto, H., & Hinson, R. E. (2023). Mobile money as a driver of digital financial inclusion. *Technological Forecasting and Social Change, 186*, 1–14.
Singh, J., Steele, K., & Singh, L. (2021). Combining the best of online and face-to-face learning:/hybrid and blended learning approach for COVID-19, post vaccine, & post-pandemic world. *Journal of Educational Technology Systems, 50*(2), 140–171. https://doi.org/10.1177/00472395211047865
Smith, K., & Hill, J. (2019). Defining the nature of blended learning through its depiction in current research. *Higher Education Research and Development, 38*(2), 383–397.
So, H. J., & Bonk, C. J. (2010). Examining the roles of blended learning approaches in computer-supported collaborative learning (CSCL) environments: A Delphi study. *Journal of Educational Technology & Society, 13*(3), 189–200.
Sun, P. C., Tsai, R. J., Finger, G., Chen, Y. Y., & Yeh, D. (2008). What drives a successful e-learning? An empirical investigation of the critical factors influencing learner satisfaction. *Computers in Education, 50*(4), 1183–1202.
Tarhini, A., Hone, K., Liu, X., & Tarhini, T. (2017). Examining the moderating effect of individual-level cultural values on users' acceptance of E-learning in developing countries: A structural equation modeling of an extended technology acceptance model. *Interactive Learning Environments, 25*(3), 306–328.
Truss, A., & Anderson, V. (2023). The navigational challenges of a blended learning approach to teaching in business and management. *International Journal of Management in Education, 21*(1), Article 100733.
Wright, B. M. (2017). Blended learning student perception of face-to-face and online EFL lessons. *Indonesian Journal of Applied Linguistics, 7*, 64–71.
Yean, L. S. (2019). Promoting active learning and independent learning among primary school students using flipped classroom. *International Journal of Education, 4*, 324–341.
Zheng, W., Yu, F., & Wu, Y. J. (2022). Social media on blended learning: The effect of rapport and motivation. *Behaviour & Information Technology, 41*(9), 1941–1951. https://doi.org/10.1080/0144929X.2021.1909140